

Master Thesis

A Challenge Set for Natural Language Inference on *but*-inferred propositions

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Abstract

This thesis creates a challenge set containing *but*-inferred propositions to probe a neural natural language inference (NLI) system’s ability in understanding *but* sentences.

Although neural NLI systems have recorded landslide success in achieving excellence results in benchmark tests, how much it understands language has remained an unclear ‘black box’. Previous studies have shown that they might have been operating on idiosyncrasies. While the conjunction *but* is extremely basic and frequent in the use of English, previous studies only probed neural NLI systems on a few functions of the word, which is far from representing the vibrant potential of the hidden information veiled behind a speaker’s use of *but*.

The present study proposes ‘*but*-inferred propositions’, which focus on the pragmatic, inferential use of *but*, and puts forward a linguistically-informed, manually-crafted challenge set that inspects the system’s behavior in processing these propositions to investigate to what extent the system understands *but* like humans do. A encoder-decoder transformers model T5 is used to perform a test on this data set.

The high failure rate in general confirmed that the system does not read the subtle implications induced by the word *but*. A majority of ‘neutral’ label, as well as similar behavior with *but* and *and* suggest that the system makes no connection between a *but* sentence and what the word *but* is inferring. The system also does not excel in processing temporal ordering of events and detecting the inferential difference of a change in sentiment. In addition, the experiment discovers some unwarranted tendencies resulting in favor of certain labels. These include an influence of annotation artifacts, for example, a higher correlation between negation cues and the ‘contradiction’ label, as well as an overarching tendency to give out ‘contradiction’ when the system is presented with reversed world knowledge. It is hypothesized that the *but* inferences are difficult for the model to learn because they are never explicit in the text. Adding *but* sentences along with their subsequent inferred propositions to the training process is expected to enrich the system’s understanding of *but*

Declaration of Authorship

I, Yan Chung Li, declare that this thesis, titled *A Challenge Set for Natural Language Inference on but-inferred propositions* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a degree degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 1 July 2022

Signed:  .

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Chapter 1

Introduction

Natural language inference (NLI) is a popular task in natural language processing (NLP), where the system needs to determine whether a hypothesis is entailed by, contradicting or irrelevant to a premise. For instance, given a premise *I have a cat*, hypothesis *I have a pet* should be regarded as *entailed*. Since NLI/RTE is able to ‘evaluate the full set of skills involved in (ungrounded) sentence understanding, from resolution of syntactic ambiguity to pragmatic reasoning with world knowledge’ (Wang et al., 2018, p.5), it is often served as a benchmark test of natural language understanding (NLU). For example, the Second Workshop on Evaluating Vector Space Representations for NLP (RepEval 2017) features an NLI competition meant to evaluate natural language understanding models (Nangia et al., 2017). This task is also included in both popular NLU benchmarks, GLUE (Wang et al., 2018) and SuperGlue (Wang et al., 2019). In recent advances of deep learning methods and gigantic language models, neural NLI systems such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019b) have achieved impressive scores, but how they operate remains unclear (Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019). More explanations about neural NLI will be detailed in Chapter 2.

Many studies have looked into how neural NLI systems learn, but not enough research was done on how they process conjunctions. Richardson et al. (2020) studied neural NLI systems’ ability to process the Boolean logic of *and*. Saha et al. (2020) investigated both Boolean and non-Boolean use of conjunctions *and*, *or* and *but*. *But* is especially interesting, because it is used to express contrasts, but logically it usually has the same function as *and* (as in both propositions *p* and *q* are true in the sentence *p but q*). In addition to Boolean use of the word *but*, Saha et al. (2020) also inspected the non-Boolean use, for example, *Fowler wrote or co-wrote **all** but **one of** the songs on album*, where **not** both elements surrounding *but* are true. This *but* induces a negation and brings more implications to NLI. However, besides the Boolean and some non-Boolean uses of *but* stated above, they have not experimented with other functions of the word.

This project investigates ‘*but*-inferred propositions’, which denote some unspoken but hint information by the use of the word *but*. In particular, this thesis looks at undone actions, sentiment, world knowledge and covert noun relations. Not only is *but* inherently inferential, the word is also inherently related to negation, since the use of *but* always hints a kind of incompatibility within the sentence. Given that “but” does not usually imply negation and that neural NLI models are found to falsely rely on lexical overlap (McCoy et al., 2019), I hypothesize that the system will fail

to understand these *but*-inferred propositions and will make wrong judgments. This thesis aims to give more insights into how a neural NLI system performs on negation (Hossain et al., 2020). Furthermore, for undone actions, it is also important for the system to understand temporal ordering of events. This thesis could contribute to the ongoing work about a neural NLI system’s ability in processing temporal information (Vashishtha et al., 2020).

A linguistically motivated challenge set is built on *but*-inferred propositions. A challenge set is a small data set that consists of a hand-designed examples to probe a system’s capacity. These examples are specifically crafted for a certain capacity and are considered *difficult* for the system to solve. In this thesis, a challenge set is created specifically for *but* sentences. Sentence pairs are constructed to investigate the model’s ability to understand the implicit contradiction between the premises and the hypotheses that are negated by the use of *but* in the sentence. Moreover, these sentence pairs are carefully altered with ‘and’-substitution, negation, different temporal ordering and semantically opposite lexicons. The purpose of these alterations is to better analyze the system’s behavior in detail.

A well-performing encoder-decoder transformer model, T5 (Raffel et al., 2020), is used to be evaluated on this challenge set, and the results are analysed for more insights into how the system works. Results show that the system performs poorly on *but*-inferred propositions. It neglects the subtle inference of *but* and fails to process temporal and sentimental information properly. The system also shows some potentially erratic idiosyncrasies that wrongfully correlate certain words with certain labels, namely negation cues with ‘contradiction’, and *continue* with ‘neutral’. Learning these *but* inferences are considered difficult for the model because they are never explicitly stated in the text. Adding these pragmatic *but* sentences and their subsequent inferred propositions to the training process is expected to enrich the system’s understanding of the use of the word *but*.

The major contributions of this paper include its attempt to cover more pragmatic uses of the conjunction *but* in the English language for an NLI task that is not studied before. It also proposes a highly controlled challenge data set with said functions of *but*. The following chapters will present the resolution of the problem in detail: Numbered (ordered) lists are easy to create:

1. Background and previous work related to neural NLI, challenge sets and functions of the word *but*. The summary will lead to the research question and hypothesis of this thesis.
2. Methodology, including the creation of the challenge data set and the design of the chosen model.
3. Results and discussions. The model’s performance will be evaluated by its failure rate, followed by a qualitative analysis.
4. Conclusion and future work.

Chapter 2

Background and related work

2.1 Natural Language Inference (NLI) and its approaches

Also known as Recognizing Textual Entailment (RTE) (Dagan et al., 2005), Natural language inference (NLI) is the task to detect the inferential relationship between texts. Given a premise and a hypothesis, the goal for an NLI model is to determine that the hypothesis entails, contradicts or is irrelevant to the premise. These relationships are labeled as ‘entailment’, ‘contradiction’ or ‘neutral’ respectively. To better illustrate the concept of entailment and the format of the data, examples from the Multi-Genre Natural Language Inference (MultiNLI) corpus, the largest benchmark database for this task developed by Williams et al. (2018), are shown in Table 2.1

As explained in Chapter 1, the conjunction *but* is inferential in nature, both semantically, as q partially negates some property of p , and pragmatically, as it reveals some presumed perception of the speaker. A capable NLI system should be able to understand *but* in both senses and make correct judgment with *but*-related inferences.

The NLI task has served as a benchmark task for research on natural language understanding (NLU). In the mid-2000’s, NLI models often used alignment-based techniques (MacCartney et al., 2008) or manually engineered features (Androutsopoulos and Malakasiotis, 2010). By releasing the Stanford Natural Language Corpus (SNLI), a large NLI dataset containing over 550K examples, (Bowman et al., 2015) has spurred new interest in the task, especially using deep-learning methods. Approaches to NLI include attention (Wang and Jiang, 2015; Parikh et al., 2016), memory (Munkhdalai

premise	label	hypothesis
At 8:34, the Boston Center controller received a third transmission from American 11	entailment	The Boston Center controller got a third transmission from American 11.
Met my first girlfriend that way	contradiction	I didn’t meet my first girlfriend until later.
8 million in relief in the form of emergency housing.	neutral	The 8 million dollars for emergency housing was still not enough to solve the problem.

Table 2.1: Examples of entailment, contradiction and neutral of an NLI task from Williams et al. (2018)

and Yu, 2017), and the use of parse structure (Mou et al., 2015; Chen et al., 2016). Neural models achieved excellent results for the task, including BERT-base models (Devlin et al., 2019) and RoBERTa-base models (Liu et al., 2019b), reaching 86.7/85.9% and 90.8/90.2% accuracies on matched-MNLI and mismatched-MNLI respectively.

Neural models have shown improvements of performance on this task. However, it is not clear what linguistic knowledge these systems have learnt. Poliak et al. (2018) and Gururangan et al. (2018) first pointed out that only classifying the hypotheses also achieves high accuracy in NLI. Poliak et al. (2018) tested a hypothesis-only model on several data sets. Although having only around 55% accuracies for both matched and mismatched MNLI development sets, the model had scores approximately 20% higher than the majority-baseline. Gururangan et al. (2018) found similar results with 53.9 and 52.3% on matched and mismatched MNLI development sets respectively when the model was completely oblivious to the premises, whereas the majority-baseline had only 35.4 and 35.2%. McCoy et al. (2019) discovered that these neural models might have operated based on fallible syntactic heuristics. They tested four models on HANS data set, which was designed to diagnose the use of such fallible structural heuristics. All models showed a strong preference for the ‘entailment’ label. These studies suggested that neural NLI systems might have relied on some idiosyncratic correlation and have not behaved in the intended fashion, hence opening up discussions about to what extent these neural models “understand” the linguistic substance in NLI.

2.2 Interpretability of NLI systems

To probe the capability and limitation of NLI models, various stress-testing data sets have been proposed. Glockner et al. (2018) tested NLI systems’ ability to process lexical inferences and found out that they are limited in their generalization ability, failing to capture many simple inferences. Geiger et al. (2018) found out neural NLI models failed to process quantifiers unless there was lexical alignment. Ivan Sanchez Carmona et al. (2018) looked at systems’ biases on specific words and discovered strong performance on unseen hypernyms but weak on unseen antonyms. The models were also often influenced by simple statistical correlations between words and training labels. Ross and Pavlick (2019) looked at verb veridicality and discovered that a neural model (BERT) showed stronger systematic bias, but was more sensitive to semantic types and syntactic properties of verbs. Ravichander et al. (2019) evaluated the models’ ability in numerical reasoning and found out that they did not significantly improve over a majority-baseline, suggesting that they did not learn to reason with quantities. Naik et al. (2018) created stress tests that consist of adversarial examples with linguistic phenomena, namely antonym, numerical reasoning, word overlap, negation, length mismatch and voice. Generally in line with previous findings, all models performed poorly in these tests. Yanaka et al. (2020) studied systematicity of monotonicity and found out that neural models performed quite well, their generalization ability was limited to cases with aligned syntactic structures. Last but not least, Richardson et al. (2020) also concluded that state-of-the-art NLI models performed poorly on different semantic phenomena. However, despite the heavy usage of conjunctions in English, there is no specific NLI dataset that tests their understanding of *but* in detail. The closest work on the word *but* is by Saha et al. (2020) where they explored both Boolean and non-Boolean usage of English conjunctions. However, they only focused on the truth value of the elements coordinated by the word *but*, but not other implicit pragmatic

inferences that can be drawn from the use of *but*.

Another closely-related work is Jeretic et al. (2020), who explored pragmatic inference and built a dataset containing pragmatic implicatures and presuppositions. Pragmatic inference looks beyond the logical relationship between isolated sentence meanings and focuses on the relation between two sentences relying on the utterance context and the conversational goals of interlocutors. Although they found that BERT learned to draw pragmatic inferences with scalar implicatures and some presupposition triggers successfully, there is still room to explore BERT’s ability in understanding pragmatic inference. To study scalar implicatures, this paper used lexical cues, for example ‘some’, ‘all’ and ‘none’ etc. to inspect systems’ behavior. For presuppositions, examples were created with some degree of lexical overlap, such as premise *Jo’s cat went* and hypothesis *Jo has a cat*. Although BERT might have successfully assigned the ‘entailment’ label, it was unknown whether this label was affected by the lexical overlap, which was observed in McCoy et al. (2019). As a result, it is still unclear whether neural NLI systems can draw inference from implied meanings beyond the surface form of texts. To further examine their findings in a neural model’s success in understanding pragmatic inference, the present thesis investigates to what extent the system understands the inference triggered by *but*.

2.3 But-inferred propositions

A *but*-inferred proposition denotes a proposition inferred for a *but* sentence to make sense. Its scope includes both the inferential element inferred by a *but* sentence and its presupposition behind. To further illustrate this, I will first revisit the meanings of *but* and the mechanics of how these meanings are constituted.

2.3.1 Inferential element

But is a conjunction that conjoins two linguistic components. Traditionally, the use of “but” is said to have two functions (Lakoff, 1971). The first is the denial of expectation. In this sense, in a *p but q* construction, the first component *p* arouses the hearer’s expectation, and the second component *q* denies it:

I am full, (Expectation: I don’t want to eat.) but I want to eat.

This adversative meaning is reported to rely more on the context, hence is often related to the study of pragmatics or discourse analysis. *But* is recognized as the “canonical” conjunction of the **adversative function** (Malchukov, 2004, p. 179).

Another use is semantic opposition (or contrast). The first clause provides a context for the second clause to show a contrast. For example:

Gavin is poor but Tom is rich.

Here, the *p* component does not necessarily arouse a certain expectation, nor it is the speaker’s intent to deny it. Instead, it poses a contrast of how *p* and *q* are different from each other. This use is referred to as **contrastive**.

However, the distinction between an adversative interpretation from a contrastive one is not always crystal clear. This is because both uses share some properties: they both negate a proposition derived from *p* with which *q* is considered incompatible (Lagerwerf, 1998; Blakemore, 1989; Umbach, 2005), and although the adversative use is claimed

	<i>p</i>	(inferential element)	but	<i>q</i>
Denial of expectation	<i>I am full.</i>	(I don't want to eat.)	<i>but</i>	<i>I want to eat.</i>
Semantic contrast	<i>Gavin is poor.</i>	(Everyone / Tom is also poor.)	<i>but</i>	<i>Tom is rich.</i>

Table 2.2: Examples of inferential elements in both uses of *but*

to rely more on the context, it is also impractical to dissect the contrastive interpretation from its context (Blakemore, 1989).

Scholars have tried to analyze the use of *but* under different theoretical frameworks to dissolve this perplexity. One approach is to assume an ‘inferential element’ derived from *p*. Different schools of study have named this concept differently, including ‘conventional implicature’ (Grice, 1975), ‘contextual assumption’ (Blakemore, 1989), ‘argumentative goal’ (Winterstein, 2012) and ‘particularized implicature’ (Geurts and Maier, 2013). No matter whether *but* is adversative or contrastive, this inferential element is always negated by *q*. Table 2.2 illustrates how an inferential element is applied in both uses of *but*.

2.3.2 Presupposition

Apart from inferential elements, *but* sentences also infer presuppositions. Presuppositions are background facts that the speaker takes for granted when producing an utterance (Beaver, 1997). For *but* sentences, presuppositions serve as the foundation on which the derivation of the inferential element is based. For instance, from a text *I am full, but I want to eat*, the inferential element *I don't want to eat* is derived from a presupposition: *When people are full, they don't eat*. This presupposition can be the speaker's subjective belief, or world knowledge. For instance, the inferential element *Everyone / Tom is also poor* from the premise *Gavin is poor* is derived from the speaker's perception that: *Everyone's / Tom's financial situation is the same as Gavin*.

Although both adversative and contrastive uses of *but* utilize inferential elements in the same fashion, (Winterstein, 2012) noticed a difference in their presuppositions. Those of the adversative use are more sensitive to the context or world knowledge, and the contrastive ones rely less on them.

To sum up, ‘*but*-inferred proposition’ is a novel, newly coined term used in this thesis to denote either an inferential elements or a presupposition that is derived by the use of *but*.

2.4 Research focus

Since the goal of NLI is to determine the underlying, unspoken inferential relationship between texts, a capable NLI system should be able to aptly process *but* sentences and their inherently inferred propositions. The present section outlines the four specific types of inferred propositions addressed in this thesis. In particular, I investigate one type of inferential element and three types of presuppositions, namely *undone actions*, *sentiment*, *world knowledge* and *noun relations*.

2.4.1 Undone actions

The inferential element under investigation is an action that is first done and then undone. Referred to as a kind of “inferred semantic opposition” by Peterson (1986) who studied children’s speech, this undoing of the action is sometimes “not directly stated but rather inferred” (Peterson, 1986, p. 585). In this example adapted from (Peterson, 1986, p. 585):

I kept the frog in a jar, but I thought it was going to die. (Inferential element: the frog continued to be kept in a jar.)

The use of *but* only makes sense when the second clause is negating some information suggested by the first clause. If no further contextual information is provided, the listener’s interpretation can only be that the action of keeping the frog in the jar has been undone, implying the frog was not kept in the jar anymore. In this case, the discontinuation of the frog being kept in a jar is not explicitly stated but inferred, thus negating the inferential element suggesting “the frog continued to be kept in a jar”. The *but* clause marks the temporal change of the state of being, continuation or existence.

Another aspect of *undone action* is that it is related to the negative sentiment. If a negative sentiment is attached to this action, the undoing of the action is implied sometimes; but never when the sentiment is positive. Consider this sentence pair:

I became a lawyer, but the work was tedious. (Inferential element: I continued to be a lawyer.)

I became a lawyer, but the work was interesting. (Inferential element: The work of a lawyer was not interesting.)

Under the circumstances where no other context is given, the listener will have very different interpretations of what the speaker is trying to convey. The first sentence can be interpreted as the speaker stopped being a lawyer, but the second sentence does not imply the same proposition. Instead, it has more to do with how the speaker *feels* about being a lawyer.

2.4.2 Sentiment

The first type of presuppositions is sentiment. When the *but* clause conveys a kind of sentiment, not only does it determine whether the action is undone, it also reveals that the speaker is presupposing an opposite sentiment, thus the incompatibility of *but* takes effect. For example:

I became a lawyer, but the work was interesting. (Inferential element: The work of a lawyer was **not** interesting.)

The use of *but* suggests that the lawyer work is not supposed to be interesting, but the speaker has changed their mind or they were surprised. The derivation of this inferential element has to come from the speaker’s negative sentiment towards being a lawyer, thus the presupposition *I thought being a lawyer would not be interesting*.

Obviously, this presupposition hinges on the sentiment expressed by the text. If the sentiment changes from positive to negative in this *lawyer* example, it discloses an opposite presupposition as well as the derived inferential element:

I became a lawyer, but the work was tedious. (Inferential element: The work of a lawyer was not tedious.)

In this case, it is implied that the speaker did not find the work of a lawyer tedious in the beginning.

2.4.3 World knowledge

The second type of presupposition is world knowledge. Some *but* sentences only make sense when the speaker presupposes some external, general world knowledge. For instance:

I am full, but I want to eat. (Inferential element: I don't want to eat.)

Here, the word *but* implies the expectation of the speaker's reluctance to eat, and the following *but* clause denies it. However, the derivation of this inferential element has to be based on a 'fact' that: *When people are full, they don't eat*, which is already understood by both the speaker and the listener. This sentence would not be considered correct if this knowledge is rebutted.

To further inspect whether the system has acquired some prior knowledge from its pretraining, we may resort to testing the system on fictionalized, twisted real world knowledge. The following examples are inspired by Winterstein (2012) and adopted from (Winterstein, 2012, p.1873):

*This ring is **cheap**, but well-made.* (World knowledge: Cheap rings are usually not well-made.)

*This ring is **expensive**, but well-made.* (World knowledge: ? Expensive rings are usually not well-made.)

Here, the use of *but* in *cheap but well-made* suggest that *cheap rings are usually not well-made*, thus the use of *but* brings an element of surprise. On the other hand, *expensive but well-made* seems to suggest an incompatibility between *expensive* and *well-made*, which goes against common knowledge. If the system indeed solely processes surface information provided by the input, its behavior should be consistent among real and fake world knowledge. Otherwise, the system may show some inconsistency when fed with real and fake world knowledge.

2.4.4 Noun relations

The last type of presuppositions is the implied relation between the nouns in a *but* sentence. In some cases, the relevance between two nouns is not explicitly expressed, but rather implied by the use of *but*. Consider the following example:

*The **internet post** was edited. People knew about **the secret**.*

Here, the relevance of *the secret* to the *the internet post* is rather opaque and open to different interpretations. However, if the two sentences are conjoined by *but*, an inferential element that put the two nouns together can be derived:

*The **internet post** was edited, but people knew about **the secret**.* (Inferential element: Upon the editing of the post, people wouldn't know about the secret.)

As such, a presupposition designating the relationship between *the post* and *the secret* can be discovered: *The secret was deleted from the post.*

2.5 Research question, hypothesis and contribution

Upon reviewing what has been done regarding the probing of the neural NLI system and the conjunction *but*, this thesis hopes to answer the research question:

Does a neural NLI system understand the pragmatic functions of *but* outlined in this chapter and make right judgments about its subsequent inference?

To investigate the system's ability in understanding *but*, this thesis tests the system on whether it differentiates between the conjunction *but* and *and*.

I hypothesize that neural NLI systems are unable to achieve human-like judgments with these propositions based on previous discoveries about NLI’s reliance on lexical overlap (McCoy et al., 2019) and prior, pre-trained linguistic knowledge (Poliak et al., 2018). The system is expected to assign a lot of ‘entailment’ if the premise and hypothesis is similar, regardless of hinted contradiction.

In addition, this thesis also hopes to resonate with recent interesting findings about neural NLI models’ ability in processing nuanced information related to *but*-inferred propositions. Firstly, this thesis also looks into negation. Hossain et al. (2020) found out that neural NLI systems struggles with negation when negation words were introduced to premise/hypothesis sentence pairs. The systems recorded an approximately 20% drop in accuracy from the original MNLI development set. Furthermore, neural NLI systems were found to be terrible at processing temporal ordering of events. This thesis also introduces negation to the data set and analyzes the model’s behavior.

Secondly, this thesis responds to previous findings about the models’ inability in processing temporal ordering of events. Vashishtha et al. (2020) inspected three models on three event order data sets and despite varying results, all of them only scored 34 - 54% accuracy, where the best performing system, RoBERTa, only managed to surpass 50% which was equivalent to the majority-baseline. This thesis also aims to provide insights into neural NLI models’ ability in dealing with negation and temporal ordering. For *undone action*, the temporal ordering is crucial to the interpretation of whether the action is undone, since the first instance has to happen first to provide the context for the second instance.

Lastly, this thesis investigates how the model interacts with world knowledge. As Glickman and Dagan (2005) argued, the notion of textual entailment is relevant only for hypothesis that are not world facts. However in Poliak et al. (2018)’s study, it has been pointed out that a hypothesis-only system might have drawn inference with only prior background knowledge of language. Therefore, it would be interesting to examine if a neural NLI system has learned this kind of underlying, experiential information. If the system does operate on some pre-acquired knowledge beyond the given input, it should show some inconsistent behavior compared to other examples.

Chapter 3

Methodology

This chapter introduces the data creation process and the model used in this thesis.^{[1](#)}

3.1 Data creation

In order to answer the research questions, the present challenge set is designed and created in a way that the system’s abilities in processing *but* sentences and each fine-grained property are systematically tested. This includes to make up *but* examples for each proposition type, and to single out specific properties to compare the system’s behavior.

[Ribeiro et al. \(2020\)](#) proposed a framework to evaluate a NLP model, which is to first discern the model’s linguistic capabilities then perform tests on each capability. The paper included three particular test types: Minimum Functionality Test (MFT), Invariance Test (INV) and Directional Expectation Test (DIR). MFT denotes a simple test case that targets a particular behavior. An INV is a test when some perturbations are applied to an example and the system output should not change. A DIR, on the other hand, should trigger a change in the prediction after the test case is perturbed. For example, if the user wants to inspect a system’s ability in sentiment analysis, they can create an MFT to see if it successfully give the right sentiment label. To inspect the same system’s robustness, they can design an INV, where examples are perturbed with noise or spelling error, and the sentiment label should not be affected by such perturbations. To inspect the system’s lexical ability, the user can use a DIR where words that convey the sentiment is replaced with one with a different sentiment (e.g., perturb ‘I love it’ to ‘I hate it’) and expect a change in the label (‘positive’ to ‘negative’).

In this thesis, MFTs and DIRs are adopted. The neural NLI model is evaluated on a set of manually crafted, labeled examples where the hypotheses are propositions inferred by the premises as MFTs. The system is directly tested on these examples to compare the expected labels and its predictions. The system’s performance and behavior serve as a reflection of what it has learned and how it processes an NLI task.

To induce more interesting discussions, each example is carefully manipulated in several ways. More refined properties of the four types of propositions: *undone action*, *sentiment*, *world knowledge* and *noun relation* are also viewed as individual linguistic capabilities of an NLI system. Each of these capabilities is singled-out and a corresponding alteration is performed on each example. The aim is to investigate the system’s

¹The challenge data set, codes for the experiment and the results can be found on <https://github.com/rod-ycli/but-inferred-propositions>

	premise	label	hypothesis
undone_action	I kept the frog in a jar, but I thought it was going to die.	contradiction	The frog continued to be kept in a jar.
sentiment	I became a lawyer, but the work was interesting.	contradiction	I thought being a lawyer would be interesting.
world_knowledge	This ring is cheap, but well-made.	contradiction	Cheap rings are usually well-made.
noun_relation	The internet post was edited, but people knew about the secret.	contradiction	The secret was added to the internet post.

Table 3.1: Base examples, one from each type of *but*-inferred propositions.

behavior in detail by observing its predictions on these tiny, highly controlled alterations. Treated as DIRs, their prediction labels are expected to be different compared to the original example. The system’s behavior reflects its capability of processing a single change in a piece of information.

Some properties and alterations are globally applied to all examples. Some are specific to certain types of *but*-inferred propositions. These properties and alterations, corresponding to the phenomena in the previous chapter, are explained in the next section.

3.2 Challenge set creation

3.2.1 Base examples (MFT)

A set of *but* sentences and inference-related propositions are created as the premises and the hypotheses. These ‘base’ examples serve as a basis for perturbation. To maintain consistency, all base examples are designed in a such a way that they are all labeled with ‘contradiction’.

When creating the base examples, the rule of thumb is to simplify the premises as much as possible to avoid other cues giving away possible inference besides *but*. Although unavoidable in some hypotheses, words like *still* and *even* in the premises may already give preference to a certain interpretation, undermining the role played by *but* and the subsequent component. Table 3.1 shows one example from each type of *but*-inferred propositions.

3.2.2 Global alterations: *and*-substitution and negation (DIR)

All examples are altered with *and*-substitution and negation.

To confirm the role of the word *but* in every example, it is substituted by *and* and the prediction is expected to change. The idea is that if the propositions are truly *but*-inferred, the substitution of *but* should also neutralize the ‘contradiction’ designed between the premise and the hypothesis, thus resulting in a change in the entailment label. For instance, when the *but* in an *undone action* is substituted by *and*:

*I kept the frog in a jar, **and** I thought it was going to die.*

	alteration	premise	label	hypothesis
undone_action		I kept the frog in a jar, but I thought it was going to die.	contradiction	The frog continued to be kept in a jar.
	and	I kept the frog in a jar, and I thought it was going to die.	neutral	The frog continued to be kept in a jar.
	negation	I kept the frog in a jar, but I thought it was going to die.	entailment	The frog did not continue to be kept in a jar.
	temporal	I thought the frog was going to die, but I kept it in a jar.	entailment	The frog continued to be kept in a jar.

Table 3.2: An example of the base and alterations of an undone action.

The undoing of keeping the frog in the jar is not implied anymore. If the system is able to understand this function of *but*, the *and*-substitution should urge the system to make a different prediction, since it no longer contradicts the hypothesis.

To inspect the system’s capability of processing negation words, another alteration is to add a negation word to the hypothesis. When the negation cue is added to the hypothesis, the entailment relationship between the premise and the hypothesis should be flipped. If the system processes negation correctly, the prediction is expected to show difference from the base example.

Examples of these two perturbations are shown in Table 3.2

3.2.3 Undone action: temporal ordering of events (DIR)

For undone actions, there is a fixed temporal ordering of p and q . Event p (action done) must precede event q in order to be ‘undone’. In the *frog* example, *but* implicitly negates the inferential element suggested by the the first clause, which is an expectation that *the frog continued to be kept in a jar*:

I kept the frog in a jar, but I thought it was going to die. (Inferential element: the frog continued to be kept in a jar.)

If the temporal ordering is jumbled by, say, swapping the position of p and q , then the action of *keeping the frog in a jar* is no longer undone but rather continued:

I thought the frog was going to die, but I kept it in a jar. (Inferential element: I kept it in a better environment.)

To inspect whether the system is able to capture the intricacy between *but* and the temporal ordering despite the exact same words used in the sentence, the positions of p and q are switched for undone actions as an alteration. Prediction is expected to change after switching. In addition to *and*-substitution and negation, Table 3.2 illustrates an example from this category of inferred propositions.

A point to note is when creating examples for this category, the verbs and the tense are carefully selected to allow for different interpretations of the action being ‘undone’ or not. For instance, saying *I kept the frog in a jar* is preferred over *the frog was in*

	alteration	premise	label	hypothesis
undone_action		I became a lawyer, but the work was tedious.	contradiction	I continued to be a lawyer.
	and	I became a lawyer, and the work was tedious.	neutral	I continued to be a lawyer.
	negation	I became a lawyer, but the work was tedious.	entailment	I did not continue to be a lawyer.
	temporal	The work was tedious, but I became a lawyer.	entailment	I continued to be a lawyer.
	sentiment	I became a lawyer, but the work was interesting.	entailment	I continued to be a lawyer.

Table 3.3: An example of the base and alterations of an undone action with sentiment.

a jar, because the verb *was* may already imply that *the frog is no longer in the jar now* without the *but* component. Meanwhile, the former sentence allows for both the changing and the unchanging of the state of ‘being kept in a jar’, depending on the second component. The past tense must therefore also be used in the hypothesis.

3.2.4 Undone action: sentiment (DIR)

In some cases when q conveys a negative sentiment, the undoing of the action is also implied. Interestingly, when the sentiment is positive, it seems that the undoing does not apply. For instance,

I became a lawyer, but the work was tedious. (Implied: ‘being a lawyer’ is undone.)

I became a lawyer, but the work was interesting. (Undoing not implied)

Therefore, another alteration where the sentiment of q is flipped is added onto the other alterations for this category. The prediction is expected to change (Table 3.3).

3.2.5 Sentiment: reversed (DIR)

The sentiment of q determines the speaker’s presupposition about the concerned topic. If q is positive, the topic of p is unfavorable to the speaker and vice versa. Since a part of undone actions are also sentiment-related, the same premises but different hypotheses are used to test the system’s capability of capturing the speaker’s sentiment towards the topic of p . An alteration is to reverse the sentiment in the premise, and the prediction is expected to change. An example for sentiment is illustrated in Table 3.4.

3.2.6 World knowledge: reversed (DIR)

As discussed in the previous chapter, *but* sentences can infer a presupposition that is accepted as world knowledge. Inspired by (Winterstein, 2012), sentences that contradict real world knowledge can be formulated to inspect the system’s ability to infer such

	alteration	premise	label	hypothesis
sentiment		I became a lawyer, but the work was interesting.	contradiction	I thought being a lawyer would be interesting.
	and	I became a lawyer, and the work was interesting.	neutral	I thought being a lawyer would be interesting.
	negation	I became a lawyer, but the work was interesting.	entailment	I did not think being a lawyer would be interesting.
	reversed	I became a lawyer, but the work was tedious.	neutral	I thought being a lawyer would be interesting.

Table 3.4: An example of the base and alterations of a sentiment presupposition. Note that the premises are taken from the undone action with sentiment, but with different hypotheses.

knowledge only from the text on the surface level. Take the following two cases as examples:

This ring is cheap, but well-made. (Presupposition: Cheap rings are usually not well-made.)

This ring is expensive, but well-made. (Presupposition: ?Expensive rings are usually not well-made.)

The latter is considered strange to human, but if it is presumed to be true, a reversed presupposition can be derived. When presented together with the same hypothesis, humans should be able to tell that the underlying presupposition changes, hence the entailment of the hypothesis also changes according to this ‘wrong common sense’. Judging only by the input, the system should also be able to logically infer this reversed presupposition if the second sentence is assumed to be true, despite violating ‘common sense’. Therefore, this reversed premise is considered an alteration for world knowledge presuppositions. To inspect whether a system’s behavior is affected by prior knowledge of the world, it will be evaluated on: 1) whether its prediction changes; 2) what label it gives to these reversed world knowledge. An example is illustrated by Table [3.5](#).

3.2.7 Noun relation

For noun relations, no additional alterations are done on top of the basic *and*-substitution and negation ones. For example, the following sentence pair poses as a base example for noun relations:

The internet post was edited, but people knew about the secret. (contradiction) The secret was added to the internet post.

And-substitution is performed on this base example to verify the system’s understanding of *but*, since the relation between *the secret* and *the internet post* sounds completely different when *and* is used instead of *but*:

The internet post was edited, and people knew about the secret. (entailment) The secret was added to the internet post.

In addition, a negation word is added to the hypothesis to see if the system is able

	alteration	premise	label	hypothesis
world_knowledge		This ring is cheap, but well-made.	contradiction	Cheap rings are usually well made.
	and	This ring is cheap, and well-made.	neutral	Cheap rings are usually well made.
	negation	This ring is cheap, but well-made.	entailment	Cheap rings are usually not well made.
	reversed	This ring is expensive, but well-made.	neutral	Cheap rings are usually well made.

Table 3.5: An example of the base and alterations of a world knowledge presupposition.

	alteration	premise	label	hypothesis
noun_relation		The internet post was edited, but people knew about the secret.	contradiction	The secret was added to the internet post.
	and	The internet post was edited, and people knew about the secret.	entailment	The secret was added to the internet post.
	negation	The internet post was edited, but people knew about the secret.	entailment	The secret was not added to the internet post.

Table 3.6: An example of the base and alterations of a noun relation presupposition.

to make a different judgment regarding the noun relation:

The internet post was edited, but people knew about the secret. (entailment) The secret was not added to the internet post.

The said examples are also presented in Table 3.6.

To conclude all the base examples and alterations, Table 3.7 shows the number of examples generated for each type of inferred proposition.

3.3 Model

One of the well-performing neural systems for NLI is Google’s Text-to-text Transfer Transformer (T5) (Raffel et al., 2020). There are two ideas behind the development of this model. The first is to fully exploit the potential of transfer learning, which is to use a model pre-trained on gigantic data and fine-tuned to perform downstream tasks. The second idea is to treat different NLP tasks as ‘text-to-text’ problems, i.e. taking text as input and producing new text as output. These tasks include sentence acceptability judgment, sentiment analysis, paraphrasing/sentence similarity, NLI, coreference res-

	Number of examples						total
	base	alteration					
		and	negation	temporal	sentiment	reversed	
Undone action	13	13	13	13	5	/	57
Sentiment	5	5	5	/	/	5	20
World knowledge	11	11	11	/	/	11	44
Noun relation	10	10	10	/	/	/	30
Total	39	39	39	13	5	16	151

Table 3.7: Number of examples (base and alterations) generated for each type of inferred proposition.

Input	Target
Thank you <X> me to your party <Y> week .	<X> for inviting <Y> next <Z>

Table 3.8: Example of inputs and targets produced by the span replacement objective applied to the input text “Thank you for inviting me to your party last week .” taken from Raffel et al. (2020). <X>, <Y>, and <Z> denote sentinel tokens that are assigned unique token IDs.

olution, sentence completion, word sense disambiguation and question answering. In short, this model is pre-trained on a huge amount of language data, and then adopts a prompt-based fine-tuning for downstream tasks.

3.3.1 Pre-training

T5 is pre-trained on the Colossal Clean Crawled Corpus (C4) developed by Raffel et al. (2020), a cleaned version of Common Crawl’s web crawl corpus which consists of hundreds of gigabytes of English text. Since the Common Crawl corpus contains a lot of noise which is not natural language, C4 undergoes a cleaning process which involves deduplication, discarding incomplete sentences, and removing offensive or noisy content. C4 contains the web extracted text from April 2019 with the aforementioned filtering. Its size is about 750 GB, and it is released as part of TensorFlow Datasets.² With pre-training, the model is able to learn linguistic knowledge of English from the distribution of the texts it reads.

3.3.2 Training objective

It incorporates a corrupted span replacement training objective, where the model is asked to give predictions to replace the corrupted span in a sentence. An example is illustrated in Table 3.8. Instead of corrupting individual tokens, multiple consecutive tokens are corrupted as a span to speed up the training process.

3.3.3 Fine-tuning

For each downstream task, the model adopts a prompting strategy to get the target output. Each input is processed as a textual prompt with a template, and the language model is asked to derive an output filling a piece of masked information probabilistically.

²<https://www.tensorflow.org/datasets/catalog/c4>

In the beginning of the text input, a ‘task prefix’ is added to hint the model about the wanted format of the output and fine-tune the model for that specific task. Since the model is exposed with enough examples for a particular task, it learns to correlate that task prefix with the targeted output, thus returning the expected format given the task prefix. For example, to ask the model to translate the sentence “Thank you very much.” from English to German, a task prefix ‘*translate English to German*’ is added in the beginning of the input. The sequence “translate English to German: Thank you very much.” is fed to the model and it is trained to output ‘*Vielen dank.*’. Different from zero-shot learning, T5 adopts transfer learning, which means when it is fine-tuned for different tasks, the parameters continue to be updated.

3.3.4 Architecture

This model uses an encoder-decoder architecture. The encoder is fed an input sequence and the decoder produces a new output sequence. The encoder uses a ‘fully-visible’ attention mask. Fully-visible masking allows a self-attention mechanism to attend to any entry of the input when producing each entry of its output. For example, when asked to perform a ‘translate English to German’ task with the sentence *Thank you very much*, fully-visible masking enables the encoder to pay attention to the whole sentence when reading the sequence.

The self-attention operations in the Transformer’s decoder use a causal masking pattern. When producing the i th entry of the output sequence, causal masking prevents the model from attending to the j th entry of the input sequence for $j > i$. In other words, the model is prevented from seeing what is coming next and is only able to access up to the current input as it produces its output during training. The decoder is used to auto-regressively produce an output sequence. That is, at each output timestep, a token is sampled from the model’s predicted distribution and the sample is fed back into the model to produce a prediction for the next output timestep, and so on. In the English-to-German translation task, when producing the first output token *Vielen*, it cannot attend to the next output token *dank*; however, it can attend to the previous prediction *Vielen* when producing *dank*. That is why this architecture can be used as a language model (LM), which can be used for next-step prediction and sequence generation (Raffel et al., 2020, p.17). Figure 3.1 visualizes the architecture of the model.

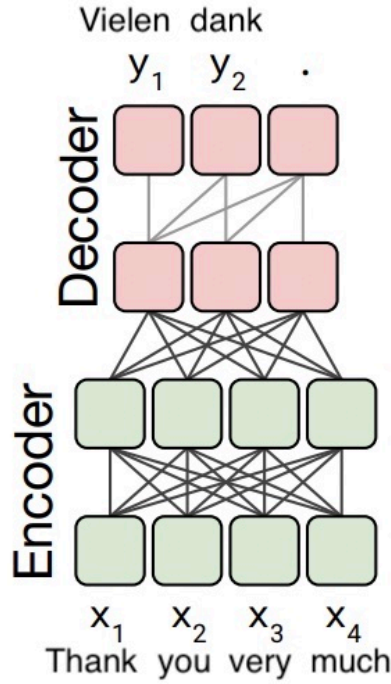
3.3.5 T5 and MNLI

For the MNLI benchmark, it is prompted with the MNLI data set, a crowd-sourced corpus with 433k sentence pairs annotated with textual entailment information. (Williams et al., 2018). An ‘*mnli*’ prefix is added in the beginning. For instance, it takes ‘*mnli premise: I hate pigeons. hypothesis: My feelings towards pigeons are filled with animosity.*’ as the input sequence. The model has observed enough examples to answer the prompt with the most probable label: ‘entailment’, ‘contradiction’ or ‘neutral’ as the output (Raffel et al., 2020, p.8).

T5 is able to outperform the previous best model and achieve state-of-the-art results for a lot of language tasks, including NLI. The model used in this paper is T5-large, with up to 770 million parameters.³ achieved 89.9 and 89.6 accuracy on the matched and

³<https://huggingface.co/t5-large>

Figure 3.1: The encoder-decoder architecture with fully-visible masking in the encoder and the encoder-decoder attention, with causal masking in the decoder taken from [Raf-fel et al. \(2020\)](#). Blocks represent elements of a sequence and lines represent attention visibility. Different colored groups of blocks indicate different Transformer layer stacks. Dark grey lines correspond to fully-visible masking and light grey lines correspond to causal masking. ‘.’ denotes a special end-of-sequence token that represents the end of a prediction. The input and output sequences are represented as x and y respectively.



mismatched MNLI benchmarks respectively.⁴ Other well-performing language models for NLI include ALBERT ([Lan et al., 2019](#)), but T5 is able to outperform ALBERT on NLI benchmarks and achieves state-of-the-art NLI results.

⁴The best performing T5 is in fact T5-11b with 11 billion parameter. Its scores are 92.2 and 91.9 respectively. However, the model is gigantic and requires way too much computing resources that are deemed not worth exploiting for the scope of this paper. Instead of insisting on using the absolute *best*-performing model, a smaller, working version with the same pretraining and architecture is adopted.

Chapter 4

Results and discussions

This chapter presents the the results of the experiment. The model’s performance is first evaluated by its failure rate in producing the expected labels. However, the failure rates alone are not conclusive of the model’s ability since they only reflect how the model performs on the examples in isolation. The model might have assigned the label due to other factors, and simple failure rates cannot demonstrate the model’s systematic, logical behavior stemming from its understanding of *but*. Therefore, an additional calculation of the model’s logical consistency is also included.

A fine-grained behavioral analysis follows.

4.1 Failure rates

Failure rate represents the percentage of the model’s failure out of all test cases. A failure is when the model do not show expected behavior. It is calculated in the following fashion:

$$\text{Failure rate} = \text{number of failed cases} / \text{number of all cases} * 100$$

For example, if there are 10 examples, and the model fails to give the expected prediction to 5 of them, then the failure rate is $5/10*100 = 50$.

4.1.1 Expected predictions

The model’s ability to process *but* sentences and draw inferences from them is reflected whether it can predict the expected results of the examples in the challenge data set. Table 4.1 details the failure rates for each proposition type: *undone action*, *sentiment*, *world knowledge* and *noun relation*, as well as their alterations: ‘*and*’-*substitution*, *negation*, *temporal*, *sentiment* and *reversed* compared with their respective expected labels.

Unsatisfactory in general

In general, the failure rates for predicting the correct expected labels for base *but* examples, as well as for all kinds of alterations, are quite high. Such poor performance is in line with this thesis’s hypothesis that the model is not capable of understanding the implicit inferences of *but*. This result is consistent with the original hypothesis that the neural NLI system is not capable of understanding *but* in these usages.

	base	alteration				
		and	negation	temporal	sentiment	reversed
undone_action	61.54	53.85	92.31	61.54	80.00	/
sentiment	80.00	20.00	100.00	/	/	60.0
world_knowledge	100.00	45.45	100.00	/	/	100.00
noun_relation	90.00	60.00	100.00	/	/	/

Table 4.1: Failure rates (%) for the four proposition types and their alterations compared with each expected label. For each example, if the model did not give the expected label, it is counted as one ‘fail’.

Better performance on *and* substitutions

Although also having fairly high failure rates, examples with the *and* alteration record much better results compared to base examples. This may prove the system’s ability in understanding *and* sentences better, and subsequent inability in differentiating the nuance between *but* and *and*.

Nonetheless, for *undone actions*, the predictor seems to have somewhat different behavior than any other types of *but*-inferred propositions. Its failure rate for base examples are lower and closer to that of *and* substitutions. This may mean the system has different interpretations regarding this particular type of propositions. This claim will be validated by a qualitative analysis in the next section.

Terrible results for negation

For negation where a negation cue is added to each hypothesis, the failure rates across all types of *but*-inferred propositions are disastrous. This result seems to be in line with Hossain et al. (2020)’s finding that transformer models struggle to make inference judgments with negation. However, it is too early to conclude that the system does not perform well with negation at all. Instead, there is a need to investigate the reason for almost 100% failure.

Poor in temporal ordering and sentiment for undone actions

Although it is far from success, the failure rate for jumbled temporal order is surprisingly not too bad at 61.54%. It means the system is able to give the correct labels to almost 40% of the examples.

On the other hand, the failure rate for *sentiment* alterations for *undone action* is as high as 80%. The system seems to be indifferent to the change of the sentiment in determining whether an action is undone.

Bad results for reversed sentiment and world knowledge

For the *reversed* type of alterations for *sentiment* and *world knowledge*, the system also failed to impress, especially for reversed world knowledge.

4.1.2 Logical consistency of alterations compared with base examples

The comparison with the expected labels alone does not justly reflect the system’s capability, because this evaluation method relies heavily on the validity of the expected

	and	negation	temporal	sentiment	reversed
undone_action	84.62	53.85	61.54	80.0	/
sentiment	80.00	20.00	/	/	40.0
world_knowledge	90.91	27.27	/	/	18.18
noun_relation	70.00	20.00	/	/	/

Table 4.2: Failure rates for logical consistency of altered examples compared with each base example.

labels. If examples are rather ambiguous or controversial, it may understate the system’s ability as to justifiably understand these texts. For instance, if ‘entailment’ is only slightly more preferable than ‘neutral’ to humans for a certain example, it is not fair to assert the system ‘failed to understand the texts’ if the prediction was in fact ‘neutral’. On the other hand, the system’s performance may be overstated when its prediction matches an alteration’s expected label, but it is actually the exact same label with the base. Since the alterations were designed in a way that the labels should change, it should not be counted as ‘success’ as the system actually failed to interpret the texts differently after the example has been altered. Instead, it reveals the system’s logical inconsistency with processing base examples and their alterations.

To further explore the system’s ability to correctly differentiate and process the alterations in a warranted fashion, an alternative method of calculating the failure rate is also included. It looks at the system’s logical consistency for alterations. For each alteration, it will be counted as failure if the label remains the same as its base, since the same label signifies the system’s failure to render a different label after blatant alterations. For example, if a base example has been falsely predicted to be ‘entailment’ and a certain alteration is also ‘entailment’, it is still a failure because it is no different than the base, even though the expected label for this alteration was supposed to be ‘entailment’. In this case, the system was being logically inconsistent with this alteration. Table 4.2 demonstrates the failure rate for each type of *but*-inferred propositions and its corresponding alterations.

Poor differentiation between *but* and *and*

The logical inconsistency for *and*-substitution is very high across all types of propositions. The fact that the labels mostly did not change after *buts* had been substituted by *ands* is in line with previous observation made in the last section that the system made much more correct predictions for *and* sentences than *but* sentences, meaning it does not differentiate *but* and *and* too well. In other words, the failure rates for getting the expected labels for ‘*and*’-substitution show that the system make more correct predictions for *and* sentences than *but* sentences. Meanwhile, the failure rates for logical consistency show that the system do not make different predictions between *and* sentences and *but* sentences. The system seems to be slightly more inconsistent with *world knowledge* and slightly better for *noun relation*.

Redemption for negation

Judging by logical consistency, the system redeems itself from the poor scores demonstrated by matching the expected labels. It means that the system is actually able to interpret the sentence pairs differently and give a different label when negation is

applied. However, the scores also indicate that there is still a good number of examples where the labels did not change after the hypotheses being negated. It could be still quite disappointing, since the ability to process negation is rather basic. However, this could be explained if the system has a preference to assign ‘neutral’ to both base and negated examples as reading both hypotheses as irrelevant to the premise, thus resulting in such logical inconsistency. Consider the following two examples:

(Premise) *I ate an apple.* (Hypothesis) *I like apples.*

(Premise) *I ate an apple.* (Hypothesis) *I **don’t** like apples.*

If both the base hypothesis and the negated hypothesis are indeed irrelevant to the same premise, it is logically consistent to give ‘neutral’ to both examples, whether there is negation. It is unfair to suggest that the system fails to process negation only because the label remained ‘neutral’.

In fact, there are 13 inconsistent cases in total (i.e. the negation label is the same as the base label), and in 10 of them both labels are ‘neutral’. Therefore, this inconsistency is not a strong proof to claim the system’s failure in processing negation. This phenomenon that both *base* and *negation* having ‘neutral’ seems to be more prevalent in *undone actions*. It may indicate that the relevancy between the sentence inputs is especially subtler for this type of propositions than others.

Poor in temporal ordering and sentiment for undone actions

The logical inconsistencies for these two types of alterations are the same as their failure rates for expected labels. This means in most cases, tampering with the event order or expressing a contrasting sentiment did not make any impact on the system’s understanding compared to the base examples.

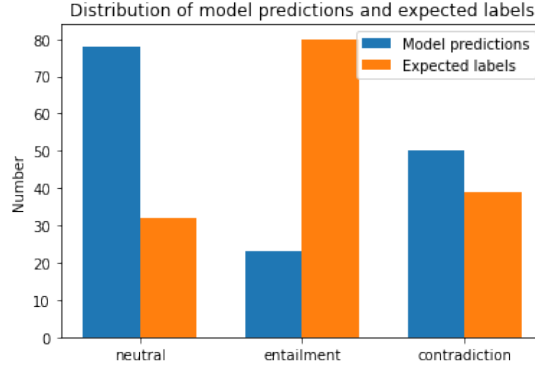
However, the model did show almost 40% success in changing the labels for temporal ordering, while it is expected to perform even worse according to Vashishtha et al. (2020)’s conclusion that language models do not learn temporal information. This may be due to the fact that Vashishtha et al. (2020)’s study uses a more fine-grained method where lexicons like ‘started’, ‘ended’ and ‘before’ are used to inspect the system’s understanding of the order of events, where the present paper only changes word order. This may exemplify that the system some degree of sensitivity to sequential ordering of the input than expected.

On the other hand, the failure rate for *sentiment* alterations reached 80%. It means that the entailment label remains the same despite the change in sentiment, confirming the system’s weakness in relating sentiment to whether the action is undone or not.

Logically consistent for reversed world knowledge

Another important finding by looking at logical consistency is that although having 100% failure in getting the expected labels, *reversed world knowledge* showed incredible logical consistency, meaning the label indeed changed when the premise was reversed. It seems to show that the system reacts strongly to the change in the lexical meaning of the premise and does not make judgments solely based on hypotheses. More discussions about world knowledge reversal and their actual label will be detailed in the next section.

Figure 4.1: The distribution of T5 large’s predicted labels and the expected labels of the *but*-inferred proposition challenge set. It shows an obvious preference for ‘neutral’ in general, where the model gave much less ‘entailment’ than expected.



4.2 Fine-grained observations and discussions

In order to discover more trends and incite more interesting discussions, the system’s behavior on each example is revisited.

4.2.1 Overall preference for ‘neutral’

Out of 151 examples in total, the system assigned 78 ‘neutral’ labels, which accounts for 51.3% of all labels, while only 32 of them are expected to be ‘neutral’. There are only 23 ‘entailment’ labels, while there are as many as 80 expected ‘entailment’s. For ‘contradiction’, there are 50 predictions for 39 expected cases. This reveals the model’s strong preference for the ‘neutral’ label. Figure 4.1 illustrates the distribution of predicted and expected labels.

Specifically, the system tends to give ‘neutral’ to base examples. Out of 39 base examples, there are 25 ‘neutrals’, which accounts for 64.1% of all base labels, while they are all designed to have ‘contradictions’. It refutes the original hypothesis that ‘entailment’ would be the prevailing label because of lexical overlap; instead, the system simply regards these base examples as irrelevant in general. In other words, despite these examples are designed how these *but*-inferred propositions are negated (‘contradiction’) by the use of the word *but*, the system simply neglects the inferred incompatibility between the premise and the hypothesis.

4.2.2 Disguised triumph for undone actions

By looking at the lower base failure rate for *undone actions*, one may assert that the system is stronger at understanding an action being undone with the use of *but*. However, out of 5 successful cases of getting the ‘contradiction’ label for these base examples, 4 of them also got the same label for their ‘*and*’-substitution alterations. This means although the system decides that the hypotheses defy the premises, it is not because of its precise understanding of the existence and the function of *but*. Instead, its improved behavior comes from other factors, perhaps the content or the phrasing of the texts itself, that lead to the contradiction, making them also contradict even with ‘*and*’-substitution. Therefore, it is not reasonable to conclude that the system works better on undone actions.

4.2.3 Possible lexical linkage to labels

There is a possible correlation between certain lexical choices and the labels. Out of 39 negation cases, only 15 of them are assigned a ‘neutral’ label and 23 of them have ‘contradiction’, while in the original design of this data set, all negation alterations are expected to be ‘entailment’. This may suggest a fallible correlation between negation words and the ‘contradiction’ label, and is in line with Gururangan et al. (2018)’s earlier findings about the relationship between negation cues and ‘contradiction’.

In addition to the linkage between negation words and ‘contradiction’, this trend is also observed with *undone action* examples. In order to contradict the ‘undoing’ of the actions taken in the premises, the hypotheses incorporated several expressions to assert the continuation of the event, including *continue doing/to do*, *kept doing* and so on. For instance:

*We were building a sand castle, but the waves came up. (contradiction) We **continued building** the sand castle.*

*I was enrolled in a private school, but I broke several school rules. (contradiction) I **kept studying** in this private school.*

There are in total 18 examples which used the word *continue(d)*, and 12 of them had a wrong ‘neutral’ label. For the word *kept/keep*, there are 22 examples in total and only 8 of them had a wrong ‘neutral’ label. This may be an indication that *continue(d)* has a stronger association with ‘neutral’ than *kept/keep*. This observation could be in line with Ivan Sanchez Carmona et al. (2018), where they found out that certain words were correlated with their training labels. It is possible that the word *continue* has a higher probability in having a ‘neutral’ label than *keep/kept* in the training data.

4.2.4 Reversed world knowledge prone to ‘contradiction’

World knowledge is an interesting proposition type to look at. It has 100% failure rate for both base and reversed examples, but only 18.18% for reversed inconsistency. By reading the predictions in detail, it is revealed that while the base examples are never labeled as ‘contradiction’ (thus the 100% failure rate), their reversed alterations are almost always ‘contradiction’.

The reason behind this is yet to be confirmed. There are two theories that may explain why reversed world knowledge is more likely to be ‘contradiction’. The first theory is that it may be attributed to the lexical antonymy presented in the premises and the hypotheses. For reversed world knowledge alterations, a certain word was replaced with its antonym to construct a ‘wrong’ common knowledge that sounds unnatural to humans. Consider this pair of examples:

Base: *It is 2 pm but it is dark. (contradiction) 2 pm should be dark.* Prediction: entailment

Reversed: *It is 2 pm but it is **bright**. (entailment) 2 pm should be **dark**.* Prediction: **contradiction**

It could be because *bright* and *dark* in the reversed example created such a conflict that the system saw it as ‘contradiction’. This is also in line with previous finding suggested in this thesis that the system does not consider the incompatibility conveyed by the word *but*. Therefore, the system might have considered the examples as *2 pm, bright* vs *2pm, dark*, thus the ‘contradiction’ label. However, when antonymy of the same kind applies to the reversed sentiment type, it did not cause such a trend in ‘contradiction’. For instance:

Reversed: *I started this diet, but I felt **terrible**. (neutral) This diet was supposed to make people feel **great**.* Prediction: **neutral**

Another possible reason is that these manipulated premises are so incompatible with real-life language use that the system predominantly decides that they are ‘contradiction’, regardless of the hypotheses. That means in the *2 pm but it’s dark* example, while *2 pm but dark* makes more sense, *2pm but bright* are much less likely to be used as an expression, thus conflicting the system’s prior understanding of language and assigned a ‘contradiction’ label. This may indicate that the system does indeed operate on some prior knowledge about the world, instead of on the semantic meaning of the texts. However, this theory does not explain the system’s behavior for all the examples. For example, the following two examples are also unnatural, but they are not predicted as ‘contradiction’:

Reversed: *The difference is **huge but important**. (entailment) Many small differences are important.* Prediction: **neutral**

Reversed: *I am **hungry, but I want to eat**. (neutral) When people are full, they eat.* Prediction: **entailment**

As for the following example, the saying *expensive but well-made* is never ‘wrong’ or ‘weird’ to humans, as it can serve as an argument in favor of *buying this ring*. However, it still got a ‘contradiction’ label, despite the premise being not exactly rare or ridiculous:

Reversed: *This ring is **expensive, but well-made** (neutral) Cheap rings are usually well made.* Prediction: **contradiction**

Therefore, the reason behind the higher tendency for reversed world knowledge to have a ‘contradiction’ is hard to conclude. Both antonymy and rarity cannot inclusively explain all similar cases. More systematic testing is needed to verify and account for the model’s biased behavior towards reversed world knowledge.

4.2.5 Reflection on the training of the model

Previous sections reviewed the model’s poor performance on predicting these *but* sentences in general. Reflecting on how T5 learns from the training process, the following paragraphs hypothesize the possible reason why the model fails to understand *but* comprehensively, and propose ways to improve its performance.

Difficulty of learning what *but* infers

It is hard for the model to fully understand why *but* is being used because the reason and its inference is never explicit in the text. The model acquires linguistic information from pre-training, when it is exposed to a huge amount of language data. Using a corrupted span objective, the model learns about language by filling in masked positions, thus obtaining knowledge about meaning and word order from the context. Although the size of C4 is so big that it covers diverse cases of *but*, the incompatibility hidden in a *p but q* construction is not explained in the text. For example, even the model has been trained with *but* sentences like *We were building a sand castle, but the waves came up* or *It is 2 pm but it is dark*, it would not know that *but* hints incompatibility, except it conjoins two clauses, which is actually equivalent to *and*.

Even when it is fine-tuned for NLI with the MNLI data set, it does not learn any additional information about *but* because MNLI examples do not require inference over

conjuncts (Saha et al., 2020). Take this instance in MNLI as an example:

*You **and** your friends are not welcome here, said Severn. (entailment) Severn said the people were not welcome there.*

The hypothesis here does not challenge the model in understanding the function of the conjunction in *you **and** your friend*; instead, it only challenges the model’s ability in generalizing them as *the people*.

To teach the model about *but*

To improve performance on NLI, the current *but* challenge set can be added to T5’s fine-tuning process. During fine-tuning, the model learns how to perform the task. It can pick up the contradiction between the *but* sentences and their inferred propositions provided in the data set, especially when they are compared with ‘*and*’-substitutions. For example:

*We were building a sand castle, **but** the waves came up. (contradiction) We continued building the sand castle.*

*We were building a sand castle, **and** the waves came up. (neutral) We continued building the sand castle.*

Then, the model is able to know the contradiction to *continue building the sand castle* changes when *but* is switched out, thus updating its understanding of the word. Although the current *but* challenge set only has a limited number of examples, Liu et al. (2019a) has shown that a model’s performance on a stress test can improve by only fine-tuning with a small number of data.

However, this may only improve the system’s performance for NLI but not other tasks. When fine-tuned on each GLUE and SuperGLUE benchmark test, parameters of the model are different for each task. To generalize and stabilize T5’s linguistic knowledge about *but* across all downstream tasks, a set of *but* sentences along with their inferences can be added to the pre-training corpus. The key is to include the incompatibility inferred by *but* explicitly so that the model can pick up from the context during pre-training. For instance:

*We were building a sand castle, **but** the waves came up. We did not continue building the sand castle.*

*it is 2pm **but** it is dark. 2 pm should not be dark.*

This is effective if alterations are also converted into sentences and are included in the corpus, especially ‘*and*’-substitution:

*We were building a sand castle, **and** the waves came up. Maybe we continued building the sand castle. Maybe not.*

*it is 2pm **and** it is dark. Maybe 2 pm should be dark. Maybe not.*

Then the model is able to contrast the use of *but* and *and*, as well as their subsequent inferences to the context.

Chapter 5

Conclusion and Future Work

This thesis inspected a neural NLI system with a challenge data set on *but*-inferred propositions. Previous studies had not exploited all the possible meanings and implications inferred by the use of *but*. To inspect a neural NLI system’s ability in understanding *but* sentences, this thesis first revisited the different functions of *but* and proposed four types of *but*-inferred propositions: undone actions, sentiment, world knowledge and noun relations. Based on previous findings about neural NLI systems’ weaknesses in relying on fallible idiosyncrasies or heuristics, this study hypothesized that the system would fail in understanding the underlying implications of *but* as humans do. However, this study was interested in finding out patterns of the system’s behavior, including on what type of *but*-inferred proposition it would perform better, and how it would process related information, namely temporal ordering, negation, change in sentiment and lexical meaning.

This thesis proposed a challenge data set, which is a linguistically-informed, small set of hand-designed examples that is considered difficult for a system to solve. A bunch of base examples were crafted to represent human understanding of each type of *but*-inferred proposition, and each base example was carefully altered to single out some fine-grained property. The most advanced NLI system to data, T5, was used for observation.

After performing a prediction task with T5, the high failure rates in general could conclude that the system failed to understand the subtle meaning conveyed by *but*. Results also suggested that the system does not differentiate the use of *but* or *and* in these examples, further proving its lack of understanding of these *but* inferences. The results for temporal alterations showed that the system does not excel in processing temporal ordering of events, but showed some sensitivity to word sequence. The system also did not show any strength in detecting the subtle inference regarding sentiment change. Other important findings include a potential correlations between certain lexicons and labels, namely a stronger tie between negation and ‘contradiction’, as well as the word ‘continue’ and ‘neutral’. Finally, there was a strong tendency for the ‘contradiction’ label for reversed world knowledge. This could be under the influence of antonymy, and the fact that the reversed examples are probabilistically rare in its training data. *But* inferences were considered difficult for the model to learn from language data because they are not explicitly stated in the text. Adding these *but* inferences, either to fine-tuning or to pre-training with explanations, could improve the model’s performance.

Future work can be done on extending the scale of the present thesis and to verify

its preliminary observations. This includes creating more examples, especially to test how the system processes real and manufactured knowledge. Although the present study hypothesizes that the system tends to give out ‘contradiction’ to reversed world knowledge, this claim needs more evidence and the reason needs to be explored. Besides, since some lexical preference for certain labels has been observed, it will also be interesting to dive deeper and unearth more arbitrary linkage between lexicons and labels. For instance, current results showed a correlation between *continue* and ‘neutral’. An extensive study containing examples alternating the use of *continue* and *keep* can be used to verify this claim. More systematic data manipulation, for example, CHECKLIST perturbations (Ribeiro et al., 2020) for masked suggestions or templates to replace different words in each example can also be adopted to examine lexical preference. As for improving the model’s understanding of *but*, future studies can try adding *but* inferences to the model’s training and see if it improves its performance.

Another aspect would be to work on the annotation task of *but* sentences. In the current study, the annotation of this challenge set is done by the author and the supervisors of this thesis, who are all good users of English. More insightful discussions about these implicit *but* inferences can be induced if an annotation task of a larger scale with more participants is performed. Annotators behavior and opinions can contribute to a clearer picture of what information the word *but* is perceived to transfer by a speaker and a listener. Moreover, insights can be elicited from a Psycholinguistics point of view about how to inspect the psychological mechanism of how humans process these *but* sentences.

The same experiment could also be done on languages. Do these inferences work for all contrastive and adversative conjunctions in different languages? Given these contrastive and adversative conjunctions have diverse pragmatic functions depending on the speaker’s intent, the context, the purpose of the discourse, the common ground shared by the participants in a conversation and different social or cultural etiquette, *but*-equivalent conjunctions are expected to play very different roles and mean very different things in different languages.

This thesis is hoped to provide more insights into to what extent a neural language model understands language, specifically the use of language. Sometimes, things are being ‘said’ beyond words. Understanding the unspoken meaning can have some applications to identifying the speaker’s intent, detecting their hidden attitudes and targeting the goal of a conversation. However, human communication is so elusive and capricious that to model it mathematically still has a long way to go. This study of *but* aims to serve as a glimpse into the constraints of current NLU systems and to contribute to future efforts to improve them.

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